S1

SYSTEM ARCHITECTURE DIAGRAM

LegalAI Resolver · Microsoft UNLOCKED

LAYER 1 — FRONTEND

index.html — Claimant Portal
- Dispute submission form
- 4-tab results dashboard
- Negotiation offer UI
- My Cases history view

respond.html — Respondent Portal
- Case summary view
- Offer submission form
- Accept / Reject proposal UI
- Secure link via email invite

HTTP / REST

LAYER 2 — BACKEND (FastAPI · Python 3.11)

- POST /api/analyze — Runs all 5 AI agents, creates case in SQLite
- POST /api/negotiate — Claimant offer → AI mediates round
- POST /api/send-invite — Sends Azure email invite to respondent
- GET /api/download-*/{id} — Download demand / settlement / court PDFs
- POST /api/respondent-offer — Respondent submits their round offer
- POST /api/my-cases — Get all cases linked to an email address

AI Calls

LAYER 3 — AI AGENT STACK

1. Intake Agent
2. Legal Agent
3. Analytics Agent
4. Document Agent
5. Negotiation Agent

Azure

LAYER 4 — AZURE COMMUNICATION SERVICES

- Transactional HTML emails
- Unique portal URL per case
- PDF download links in emails
- 6 automated email triggers

INPUT: Plain English dispute description

OUTPUT: Settlement PDF + Court File PDF

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MULTI-AGENT PIPELINE & NEGOTIATION FLOW

5-AGENT SEQUENTIAL PIPELINE — Each agent receives prior agent's full output as context

1 INTAKE AGENT
- Classify 12 dispute types
- Extract key facts
- Detect severity (LOW–CRIMINAL)
- Monetary vs non-monetary
- Structured JSON output

2 LEGAL AGENT
- Map applicable statutes
- Jurisdiction detection
- Section numbers + relevance
- Claimant rights
- Respondent defenses

3 ANALYTICS AGENT
- Win probability 0–100%
- Court fee estimate
- Timeline projection
- Confidence level
- Settlement range min/opt/max

4 DOCUMENT AGENT
- Demand Letter PDF (Navy)
- Court File PDF (Red)
- Settlement Agreement (Green)
- 3 sub-agents with own prompts
- Unicode-safe rendering

5 NEGOTIATION AGENT
- Per-round offer analysis
- Fair amount proposal
- Gap analysis + reasoning
- Pressure points both sides
- Settlement text generation

INPUT: Plain English dispute text

OUTPUT: Settlement PDF / Court File PDF

3-ROUND AI MEDIATION FLOW — Triggered after both parties submit offers each round

CLAIMANT SUBMITS OFFER — **RESPONDENT SUBMITS OFFER** — **AI MEDIATOR ANALYZES** — **AI PROPOSES FAIR AMOUNT** — **BOTH PARTIES DECIDE**

BOTH ACCEPT → SETTLED
Settlement PDF generated + emailed

REJECT (Round 1 or 2) → NEXT ROUND
New offers, max 3 rounds

REJECT (Round 3) → ESCALATED
Court File emailed to both

S3

DATA MODEL & DOCUMENT SPECIFICATIONS

SQLite — cases TABLE

case_id	TEXT PK	8-char uppercase hex
claimant_email	TEXT	Case owner email
respondent_email	TEXT	Invited party email
status	TEXT	ANALYZED / INVITED / NEGOTIATING / SETTLED / ESCALATED
created_at	TIMESTAMP	Auto-generated on insert
case_data	TEXT JSON	Full nested payload (see below)

case_data JSON PAYLOAD STRUCTURE

```
{ "case":   { dispute_type, claim_amount, severity,
is_monetary }
  "legal":  { applicable_laws, legal_standing, defenses }
  "analytics": { win_probability, settlement_range }
```

DEMAND LETTER

AI Agent

_demand_letter_agent()

Prompt Style

5-paragraph legal narrative

PDF Theme

Navy blue header & accent

Filename

demand_letter_{case_id}.pdf

CTA

7-day payment deadline

Trigger

Generated at case analysis

COURT-READY FILE

AI Agent

_court_file_agent()

Prompt Style

Structured JSON litigation pack

PDF Theme

Red header, dark sections

Filename

court_file_{case_id}.pdf

Sections

Evidence checklist, Filing steps

Trigger

Upfront + on-demand download

SETTLEMENT AGREEMENT

AI Agent

_settlement_agent()

Prompt Style

9 numbered legal clauses

PDF Theme

Green header, signature blocks

Filename

settlement_{case_id}.pdf

Sections

Payment Terms, Release, Confid.

Trigger

Only on BOTH ACCEPT

SUPPORTED DISPUTE TYPES (12): Rental Deposit · Unpaid Salary · Consumer Fraud · Contract Breach · Physical Assault · Harassment · Defamation · Property Damage · Neighbor Dispute · Domestic Matter · Apology Demand · Other

EMAIL NOTIFICATION FLOW & API REFERENCE

AZURE COMMUNICATION SERVICES — 6 AUTOMATED EMAIL TRIGGERS

1. CASE INVITE

Trigger: /api/send-invite called

To: Respondent · [Action Required] Legal Dispute Case #XXX

CTA: View Case & Respond

2. OFFER RECEIVED

Trigger: /api/respondent-offer called

To: Claimant · [Round N] Other party submitted offer

CTA: Continue to submit your offer

3. AI PROPOSAL

Trigger: /api/negotiate completes

To: Respondent · [Round N] AI Settlement Proposal

CTA: View & Decide (Accept/Reject)

4. NEXT ROUND

Trigger: Proposal rejected, round < 3

To: Respondent · [Round N] Submit Updated Offer

CTA: Submit Round N Offer

5. SETTLEMENT

Trigger: Both parties Accept

To: Both · Settlement Confirmed

CTA: Download Agreement PDF

6. ESCALATION

Trigger: Round 3 rejected by either

To: Both · Mediation Concluded

CTA: Download Court File PDF

COMPLETE API ENDPOINT REFERENCE (16 endpoints)

METHOD	ENDPOINT	DESCRIPTION
GET	`/`	Serve index.html claimant portal
GET	`/respond/{case_id}`	Serve respond.html respondent portal
POST	`/api/analyze`	Run 5 AI agents, create case
POST	`/api/send-invite`	Send Azure email invite
GET	`/api/case-for-respondent/{id}`	Get case data for respondent
POST	`/api/respondent-offer`	Respondent submits round offer
POST	`/api/negotiate`	Claimant offer → AI mediation
POST	`/api/proposal-response`	Accept or Reject AI proposal
GET	`/api/proposal-status/{id}/{r}`	Poll both parties' decisions
POST	`/api/settle`	Manual settle fallback
POST	`/api/my-cases`	Get all cases by email
GET	`/api/download/{case_id}`	Download demand letter PDF
GET	`/api/download-settlement/{id}`	Download settlement PDF
GET	`/api/download-court-file/{id}`	Download court case file
GET	`/api/case/{case_id}`	Get full case data (debug)
GET	`/health`	Health check endpoint